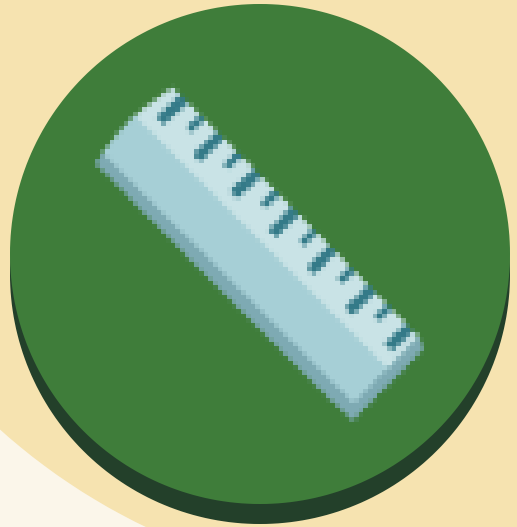
Small Engines Overhaul · Unit 1 · Level 2 (modified)

Measuring Tools



TODAY'S PLAN

- 📏 Rulers and inches
- 🗜️ Dial calipers
- 🔩 Micrometers
- 📄 Feeler gauges and specs

Review · 5 minutes

Quick Review



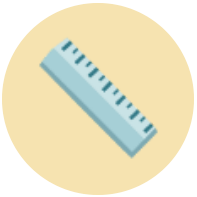
Name 3 pieces of PPE.

Where do oily rags go?

What do you disconnect before working on an engine?

Words to Know

Words to Know



Ruler

A straight tool marked in inches or millimeters to measure length.



Dial caliper

A sliding tool with a dial that measures outside, inside and depth to 0.001 inch.



Micrometer

A screw tool that measures very exactly, to one thousandth of an inch (0.001).



Feeler gauge

A set of thin metal blades of known thickness used to measure small gaps.

Words to Know

More Words to Know



🎯 **Specification**: The correct size or number the maker says a part should be (the 'spec').

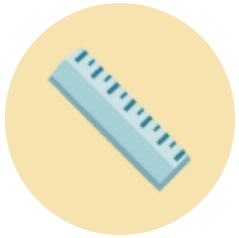
🔢 **Thousandth**: One thousandth of an inch, written 0.001. Engine parts are measured this small.

⭕ **Diameter**: The distance straight across a circle through the middle, like across a crankshaft journal.

🕳️ **Cylinder bore**: The round hole in the engine block where the piston goes up and down.

Learn

Precision Measuring Tools



Ruler

length to 1/16 inch



Dial caliper

outside, inside, depth to 0.001



Micrometer

outside size to 0.001



Feeler gauge

gaps, blades of known size

Learn

Reading a Dial Caliper



Read the **inches and tenths** on the bar (0.100 each).

Add the **dial** reading (thousandths).

Example: 0.500 + 0.025 = **0.525 inch**.

🃏 Check 1 of 2



Bar reads 0.300, dial reads 0.012. The size is…

Hold up 1, 2, or 3:

1️⃣ 0.012 inch

2️⃣ 0.312 inch

3️⃣ 3.12 inches

✅ Answer · Card 1 of 2



0.312 inch

Add bar + dial.

Learn

Reading a Micrometer



Close the spindle gently with the **ratchet**.

Read the **sleeve** (0.025 per line) and add the **thimble** (0.001 per line).

Example: 0.850 + 0.025 + 0.007 = **0.882**.

Learn

Specs and Wear



The repair manual gives the **specification** (spec).

Measure and compare: too small or too large means **worn**.

Write measured and spec side by side.

🃏 Check 2 of 2



A crankpin spec is 0.873; you measure 0.868. The part is…

Hold up 1, 2, or 3:

1️⃣ Perfect

2️⃣ Too big

3️⃣ Worn (too small)

✅ Answer · Card 2 of 2



Worn (too small)

Smaller than spec = wear.

Learn

Feeler Gauges



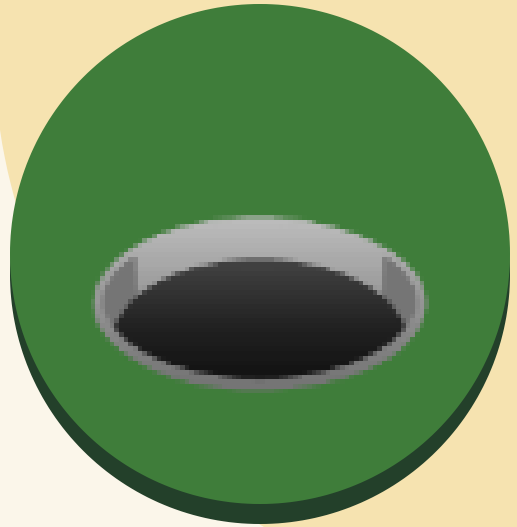
Slide blades into a gap until one fits with light drag.

Used for spark plug gap, ring end gap, valve clearance.

Never force a blade.

Steps

Measuring the Cylinder Bore



1. Clean the cylinder
2. Measure at top, middle and bottom
3. Measure in two directions
4. Compare to the spec for wear or out-of-round

Wrap-Up

Recap



Caliper
bar + dial



Micrometer
sleeve + thimble



Feeler gauge
gaps



Spec
compare to manual

Wrap-Up

Exit Ticket



Bar 0.200 + dial 0.045 = ?

What tool would you use for a spark plug gap?

What does 'spec' mean?